

The biodiversity we ignore: a global meta-analysis on taxonomic bias

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ABSTRACT

As biodiversity decline continues, there is a critical need to document, monitor, and protect species equitably across the tree of life. However, both scientific research and societal perceptions of biodiversity are often biased toward large, colorful, charismatic, familiar, or otherwise “useful” species. We conducted a global meta-analysis encompassing 905 effect sizes from 78 studies across three decades of research assessing drivers of taxonomic bias. Positive biases were widespread across study designs, regions, and taxonomic scopes, with plants and invertebrates typically receiving disproportionately less attention than vertebrates. However, the magnitude of bias was generally small (Pearson's $r < 0.5$), suggesting mild impacts on research agendas. Concerningly, conservation priority emerged as the only dimension not consistently associated with positive bias, suggesting a misalignment between research attention, funding, public interest, and conservation needs. We conclude that while biases are pervasive across taxonomic groups, they are subtle – but their conservation consequences warrant urgent correction.

INTRODUCTION

In the span of a century, humans have altered the face of the biosphere – largely for the worse (Barnosky & Hadly 2026). As the natural world erodes at unprecedented speed, we are not only losing biodiversity (Keck *et al.* 2025; Shaw *et al.* 2025), but also the collective knowledge of nature (Jaric *et al.* 2022; Soga & Gaston 2016) and even shared baselines of what healthy ecosystems look like (Soga & Gaston 2018). Yet the precise contours of this crisis remain contested, leaving ample room for debate. Can the ongoing decline in biodiversity be described as a "sixth mass extinction" (Wiens & Saban 2025)? Are extinctions occurring at comparable rates across the different branches of the tree of life (Cowie *et al.* 2022; Regnier *et al.* 2015)? And, perhaps most pressingly, do we even possess the scientific knowledge required to answer these questions with confidence?

Central to this discussion is the realization that both scientific and societal knowledge of biodiversity are profoundly fragmented (Mammola *et al.* 2023b, a; Troudet *et al.* 2017). In two seminal papers published in 2022, it was shown that research in ecology, behaviour, and conservation is dominated by studies on birds and mammals, while less 'popular' groups like amphibians and reptiles receive far less attention – phenomena described as "taxonomic chauvinism" (Bonnet *et al.* 2002) or "taxonomic bias" (Clark & May 2002). These early insights catalyzed a broad research agenda aimed at documenting and quantifying biases in biodiversity knowledge. Researchers soon recognized that this bias, intentional or otherwise, was far greater when shifting the focus beyond vertebrates. Entomologists joined the effort, showing that animals without backbones frequently fall below the radar of research funding and policy agendas – a pattern described as "institutional vertebratism" (Leather 2013). In parallel, plant scientists introduced the concept of "plant blindness" (Balding & Williams 2016; Wandersee & Schussler 1999) to capture the systematic neglect of plants relative to animals – a similar bias may be even stronger for fungi (Goncalves *et al.* 2021). At the

microbial scale, knowledge fragmentation is most extreme, with half of all bacteriological papers focusing on just ten species (Jensen 2025).

As this literature accumulates, the impression is that nearly every field can document biases affecting its own focal taxa. But is this feeling justified? How can we best navigate this growing body of evidence and take action to level the playing field? To address these questions, we present a global meta-analysis of 78 studies (905 effect sizes) spanning three decades of research on scientific and societal taxonomic bias. Using meta-regressions, we explored: (1) the prevalence and magnitude of species-level bias in biodiversity studies; (2) whether intrinsic species attributes (e.g., size, color) and extrinsic factors (e.g., extinction risk, utilitarian value) drive observed patterns; and (3) whether bias magnitude varies across outcomes (research intensity, funding, societal attention), taxa, geographical regions, and study designs. We confirmed that widespread but modest taxonomic bias favors vertebrates over other groups (most effect sizes $r < 0.5$). Publication bias inflates some reported effects, particularly in smaller studies. Most concerning, conservation priority does not predict research attention – efforts are not systematically directed toward species at greatest risk.

MATERIAL & METHODS

Literature search

We systematically collected studies investigating disparities in biodiversity knowledge using the Web of Science database. To ensure a comprehensive and reproducible literature search, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) workflow (Moher *et al.* 2009; Page *et al.* 2021) ([Figure S1](#)). We used the following searching string:

ALL = ("species" OR "organism*" OR "taxon" OR "taxa" OR "Tree of Life" OR "biodiversity" OR "nature" OR "marine" OR "terrestrial" OR "ecosystem" OR "habitat" OR "biome" OR "animal*" OR "plant*" OR "fungi" OR "bacteria" OR "virus" OR "prokaryot*" OR "eukaryot*" OR "microorg*" OR "wildlife") AND ALL = ("cognitive bias" OR "research bias" OR "vertebratism" OR "plant blindness" OR "awareness disparity" OR "favoritism" OR "prejudice" OR "judgement bias" OR "affective state") AND ALL = ("conservation" OR "ecology" OR "funding" OR "project*" OR "popularity" OR "salience" OR "gbif" OR "IUCN" OR "occurrence*" OR "knowledge" OR "research intensity" OR "research effort" OR "extinction risk" OR "citation*" OR "body size" OR "altmetric" OR "range size" OR "color" OR "attractiv*" OR "beauty" OR "aesthet*" OR "charism*" OR "survey" OR "questionnaire" OR "classroom" OR "curricul*" OR "education").

The initial search on November 30, 2023, returned 1,541 articles, which we screened for eligibility. We also set up a Web of Science alert so that new articles meeting the search criteria were sent to us by e-mail. The extraction of additional literature through the Web of Science alert continued until the end of 2025, resulting in an additional batch of 11 papers.

Inclusion criteria

We included studies that met the following criteria: (1) focused on species-level biodiversity across any habitat, ecosystem, biome; (2) quantified any form of awareness disparity, perception, research, and conservation bias towards any given biodiversity component; (3) measured bias quantitatively, for example through proxies of research interest (e.g., citations, number of papers focusing on a given taxon), proxy of research intensity (e.g., number of IUCN assessment, availability of species data and information), or proxies of people's awareness (e.g., volume of online searches for a given taxon, human preferences

as measured through surveys). We excluded studies that (1) focused on bias at the landscape and/or ecosystem level (e.g., a study showing that research on wetlands received more funds than those on groundwaters); and (2) lacked quantitative estimates (e.g., opinion papers, literature reviews).

To assess the effectiveness of our inclusion criteria, we ran inter-rater agreement tests between two authors who independently scored 100 studies. We calculated the agreement level with Cohen's kappa, which yielded an excellent level of repeatability (Cohen's kappa: 0.89; 95% confidence interval [CI]: 0.82–0.95).

Data extraction

Following the screening phase, for each paper that met our inclusion criteria we extracted predictors and response variables, all quantitative data that would enable us to calculate our effect sizes, and relevant metadata for meta-regression. It is important to note that several studies reported multiple measurements that were suitable for inclusion in the meta-analysis.

In addition, to collecting bibliographic data for each study, we recorded the geographic scope of studies (breaking down at the continent level), the taxonomic focus, and the type of study, dividing between survey-based (e.g., a study investigating public perception of biodiversity via a questionnaire) and data-mining studies (e.g., a study comparing Wikipedia page visualization in vertebrates versus invertebrates).

We captured taxonomic bias statistics aligned with four possible study designs (i.e., dependent variables): 1) Attitude, namely studies of bias with respect to the emotional response of people and scientists towards different species; 2) Research, namely studies looking at whether certain taxa are more intensively researched than others (often measured through bibliometric indices); 3) Popularity, namely studies looking at whether certain taxa are more popular than others (often measured through internet-derived metrics such as

Wikipedia views, Google Trends data, and similar); and 4) Funding, namely studies looking at whether certain taxa receive more investments for research or conservation. Given the limited sample size and preliminary analyses showing mostly positive effect sizes across these four categories (all reflecting the same underlying construct of taxonomic bias), we pooled them for the primary analysis. However, we tested whether bias type moderated the relationship with independent variables in secondary analyses (see "Data analysis").

We further categorized five main independent variables, mostly measured along a gradient or contrasted against a control group: 1) Appearance, namely whether the driver of bias is some proxy of species' aesthetic appeal or charisma; 2) Commonness, namely whether the driver of bias is some measure of how common the species is (whether geographically, ecologically, culturally, and so on); 3) Conservation, namely whether the driver of bias is the fact that a species is deemed important from a conservation standpoint (e.g., it is threatened with extinction); 4) General bias, namely analyses comparing whether a given taxon receives a biased attention, with respect to some proxy measures, compared to a control group; 5) Utilitarianism, namely whether the driver of bias is the fact that humans either use the species (e.g., an edible plant) or are threatened by it (e.g., a venomous snake). In our meta-analytical approach, we treated these variables as independent measures from one another, and hence we separated the analysis at this level (see "Data analysis"). Within each of these five categories of independent variables, we further classified more granular subgroups (Table 1), which we later explored as moderators within each main variable.

Effect size calculation

We calculated effect sizes for each relevant comparison reported within a study, including analyses involving multiple explanatory features or different factor levels. Specifically, we

converted all test statistics describing associations between proxy measures of research or awareness disparity for biodiversity components and their potential explanatory descriptors into Pearson's correlation coefficient (r), using standard conversion formulas (Lajeunesse, 2013). Pearson's r is a commonly used effect-size metric that quantifies the strength and direction of a linear association between two variables. It ranges from -1 to 1 , with positive values indicating positive associations, values near zero indicating weak or no association, and negative values indicating negative associations. We selected this metric over others because most studies in our dataset were expected to report correlative relationships between continuous variables. For studies reporting associations using chi-square (χ^2) tests without standardized effect sizes, we derived correlation-equivalent measures from the reported χ^2 statistics and sample sizes (N). When χ^2 tests had one degree of freedom, effect sizes were calculated as the phi coefficient ($\phi = \sqrt{[\chi^2/N]}$), which is equivalent to Pearson's r . When tests had more than one degree of freedom, we calculated Cramér's V ($V = \sqrt{[\chi^2/(N \times df)]}$), as ϕ (and thus r) is not defined for higher-dimensional contingency tables. We treated all resulting effect sizes as correlation-equivalent measures of association.

Data analysis

We conducted analyses in R version 4.4.1 (R Core Team 2024), primarily through the *metafor* package version 4.8-0 (Viechtbauer 2010).

Meta-analysis

We investigated patterns of bias in biodiversity knowledge and perception using meta-analytic linear mixed-effects models. In this framework, positive parameter estimates indicate a positive bias, negative estimates indicate a negative bias, and estimates close to

zero indicate no general bias. To meet assumptions of normality, we transformed Pearson's r to Fisher's z values prior to analysis (Rosenberg *et al.* 2013). For reporting and visualisation, we back-transformed model estimates and their 95% confidence intervals to Pearson's r . We considered estimates statistically significant when their 95% confidence intervals did not include zero.

As we interpreted the five predictors of taxonomic bias (Table 1) to be independent from one another, we fitted five base meta-analytic linear mixed-effects models to quantify the overall magnitude of reported disparities in biodiversity knowledge and perception across all estimates belonging to a given predictor category. To account for non-independence arising from multiple effect sizes extracted from the same publication, we included publication identity as a random effect. We then fitted a second model for each predictor category that included the more granular classification (subgroups within each predictor; Table 1) as a moderator. Subsequently, we examined the influence of several other moderators – variables expected to condition effect sizes – on these overall effects: study design (survey-based versus data-mining studies), geographic scope, focal taxonomic group, and dependent variable type (Attitude, Research, Popularity, or Funding).

Publication bias

We assessed potential publication bias by examining funnel plot asymmetry, small-study effects, and time-lag bias. Because we often extracted multiple effect size estimates from the same study, all analyses accounted for within-study dependence. Throughout, we emphasize that these analyses test for patterns consistent with small-study effects and time-related biases, but cannot unambiguously distinguish publication bias from other sources of heterogeneity among studies.

First, we visually inspected a funnel plot based on study-level aggregated effect sizes. For this, we averaged effect sizes and corresponding sampling variances within studies to obtain a single estimate per study, thereby avoiding pseudoreplication due to multiple estimates from the same source. We interpreted asymmetry in the funnel plot cautiously as indicative of small-study effects rather than definitive evidence of selective publication (Sterne *et al.* 2011).

Second, we formally tested for small-study effects using a multilevel Egger regression implemented within a multivariate meta-analytic framework. Specifically, we regressed effect sizes on their standard errors, with study identity included as a random intercept. We interpreted a significant positive association between effect size and standard error as evidence for small-study effects (Egger *et al.* 1997).

Third, we tested for time-lag bias by fitting a multilevel meta-regression with publication year as a moderator (standardized to ease model convergence and interpretation). A significant temporal trend would indicate systematic changes in reported effect sizes over time, consistent with early exaggerated findings or delayed publication of smaller effects.

RESULTS

A general overview of the research field

Our systematic literature search identified 78 studies, out of 1541 screened, that met our inclusion criteria (Figure S1), yielding 905 effect-size estimates for analysis (mean $\pm$ s.d. = 11.6 ± 36.1 effect sizes per study). Bias in biodiversity knowledge and perception was examined across a wide range of time frames (Figure 1A), geographic regions (Figure 1B), taxa (Figure 1C), and study designs (Figure 1D). The included studies were published over a 27-year window (1998–2025), with most appearing in the last decade, and spanned 54

journals across the expected broad fields of interdisciplinary research, ecology, and conservation, as well as several more specialized journals focused on education, psychology, or specific taxa. The vast majority of estimates concerned animals (72%), comprising 68% vertebrates and 4% invertebrates, followed by plants (15%) and studies including multiple distantly related taxa (11%). Studies focusing exclusively on fungi or microorganisms were exceedingly rare; these groups were considered only in combination with other organisms.

Global disparities in biodiversity knowledge and perception

Meta-analysis across the full dataset revealed a statistically significant positive bias in biodiversity knowledge and perception for measures of general bias (Pearson's $r = 0.40$; 95% CI: 0.10–0.61), a smaller but still significant effect for measures of appearance ($r = 0.18$; CI: 0.08–0.27), commonness ($r = 0.17$; CI: 0.07–0.26), and utilitarianism ($r = 0.15$; CI: 0.02–0.29), and no effect for conservation ($r = 0.02$; CI: -0.01–0.07). While most effect-size estimates clustered around zero, all distributions of effect sizes were right-skewed, with a long tail of predominantly positive values, indicating that bias is widespread but generally modest in magnitude (Figure 2A). More nuanced patterns emerged when zooming in within these independent variables (Figure 2B). All measures of appearance were positively associated with bias (slightly less for studies looking at less pronounced colorations or general appeal and charisma of species). Likewise, the commonness of species, whether cultural, taxonomical, geographical, or ecological, was consistently positively associated with bias. More variability emerged with respect to study targeting general bias. These are studies that typically compare whether a dependent variables (e.g., funding) is distributed for a given taxon compared to a control, often with very large contrasts (e.g., plants versus animals, ectotherm versus endotherm vertebrates, or invertebrates versus vertebrates). The

strongest positive effect emerged for plants ($r = 0.55$; CI: 0.26–0.75) and invertebrates ($r = 0.42$; CI: 0.09–0.67), whereas all other comparisons were not statistically significant (but also had generally low sample sizes).

The overall pattern of a widespread prevalence of bias was mostly consistent across study designs (Figure 3A), although survey-based studies tended to yield larger effect sizes than data-mining approaches (except for appearance, where no pattern was detected). Effect sizes also varied more markedly depending on how bias was operationalized in terms of dependent variables (Figure 3B). Bias estimates were generally larger when bias was measured in relation to popularity, research effort, and attention compared to funding (which was often not significant), and when independent variables captured aspects of appearance or commonness. Conversely, effect sizes were comparable across geographic regions, with some variability but no strong regional differences in the magnitude of detected bias (Figure 3C). Finally, greater heterogeneity emerged with respect to the taxonomic scope of studies, at times attributable to varying sample sizes (Figure 3D). For measures of appearance, studies on vertebrate groups tended to report larger bias estimates than studies restricted to invertebrates or those encompassing multiple taxonomic groups. By contrast, the effect of commonness was consistently positive across all taxa, showing no such taxonomic variation. Other predictors exhibited more variable or negligible patterns.

Bias in studies on bias

Visual inspection of the study-level funnel plot suggested asymmetry consistent with small-study effects, whereby smaller studies tended to report larger positive estimates (Figure S2). In contrast, larger studies showed more moderate and less variable effects, suggesting that the overall mean effect may be inflated by smaller studies. This pattern was confirmed by a multilevel Egger regression, which revealed a strong positive association between effect size

and standard error ($\beta \pm SE = 1.88 \pm 0.09$, $p < 0.001$). Importantly, the extrapolated effect size for larger studies remained positive, although substantially smaller in magnitude.

In contrast, there was no evidence for time-lag bias, as effect sizes did not systematically vary with publication year ($\beta \pm SE = -0.0005 \pm 0.06$, $p = 0.99$). This suggests that the observed asymmetry is unlikely to be driven by early exaggerated findings and instead reflects persistent differences between smaller and larger studies.

DISCUSSION

This meta-analysis provides a field-wide view of the literature examining biases in biodiversity knowledge and perception. The body of work we reviewed is global in scope, encompassing a broad range of taxa, geographic regions, and methodological approaches. Yet even here, a familiar pattern emerges: the studies themselves exhibit the very taxonomic bias they aim to document. Vertebrates dominate the meta-literature just as they dominate the primary literature (Figure 4), while entire kingdoms – particularly fungi and microorganisms – remain largely absent, even from research explicitly designed to expose such gaps. Perhaps most surprisingly, this pattern proved remarkably uniform across continents. Despite differences in regional biodiversity, research traditions, and conservation priorities, we found no major geographic variation in bias studies – the same groups are neglected everywhere. Importantly, however, the Global South remains largely underrepresented in taxonomic bias studies, highlighting that probably our current knowledge could be further improved (e.g. see Ondo *et al.* 2024).

Our intuition that some form of bias is detected whenever one looks for it was largely confirmed. Positive biases – reflecting greater research attention, knowledge, funding, or interest toward particular taxa or traits – were widespread across study designs, regions,

and taxonomic scopes (Figure 2). However, the magnitude of these effects was generally small, with most standardized effect sizes well below $r = 0.5$. This indicates that while biases are not exceptional anomalies but pervasive, they are typically modest in size rather than overwhelming (Figure 3). The system, it seems, is not fundamentally broken – just quietly and persistently skewed.

A result that may seem counter-intuitive is that for measures of appearance, studies on vertebrates – birds, amphibians, reptiles, and mammals, but not fish – tended to report larger bias estimates than studies restricted to invertebrates or multiple taxonomic groups. This suggests that within vertebrate clades, disparities between the most and least charismatic species are particularly pronounced (e.g., Frynta et al. 2013, dos Santos et al. 2020, Santangeli et al. 2023). A resplendent quetzal commands far more attention than a drab sparrow; the gap in attention between the most and least visually striking vertebrates appears wider than comparable divides among invertebrates or plants.

From this perspective, our findings are cautiously reassuring. If biases in biodiversity knowledge are widespread but generally mild, they do not imply a catastrophic distortion of the global research agenda. This pattern was particularly evident for research funding: although some studies have reported taxonomic biases in the allocation of conservation and research funding (e.g., Negron-Ortiz 2014, Mammola et al. 2020, Adamo et al. 2022), aggregate effect sizes in our synthesis were comparatively weaker and sometimes indistinguishable from zero. An important caveat is warranted, however. A recent large-scale analysis of approximately 14,600 species-based conservation projects over a 25-year period reported stronger and more widespread taxonomic biases in funding than previously documented (Guenard et al. 2025). This study could not be included in our meta-analysis because its analyses were largely descriptive, precluding the extraction of standardized effect sizes, but its conclusions nonetheless suggest a more concerning scenario.

Another result of particular relevance to biodiversity conservation is that conservation priority emerged as the only dimension not consistently associated with positive bias (except, curiously, in Oceania). In other words, research attention, funding, and public interest are not systematically aligned with species or groups facing the greatest conservation risk. Charisma, visibility, and familiarity continue to shape attention more strongly than vulnerability or data deficiency. This suggests that sometimes, entering the political arena to influence conservation decisions is more effective than accumulating data (Ellison 2016).

Importantly, our results also show that estimates of taxonomic bias reported in the literature are themselves subject to bias. Publication-bias analyses revealed evidence of small-study effects, whereby smaller studies tended to report larger effect sizes (Figure S2). Many of these studies relied on surveys, questionnaires, or classroom-based designs, where sample sizes are inherently constrained and effect estimates more variable. Conversely, extrapolations from multilevel models indicated that effect sizes for larger, often data-intensive studies remained positive, albeit substantially smaller. In other words, the signal appears robust, but its magnitude may be inflated in parts of the literature.

OUTLOOK: CHANGING THE *STATUS QUO*

Taken together, our results suggest that biases in biodiversity knowledge and perception are real, pervasive, and structural, but also more nuanced than often portrayed. Rather than dictating the direction of biodiversity research outright, these biases appear to shape its margins, interacting with feasibility, methodology, historical research trajectories, and institutional constraints. So, we must confront the important question: how do we improve on the *status quo*?

If research is merely skewed, and not broken, we do not need a systemic revolution — but we do need deliberate, targeted action if biodiversity research is to expand beyond the usual suspects. For conservation, given the ubiquitous taxonomic bias and the challenge of correcting it, a promising path forward may be to redirect focus toward habitats (Adamo *et al.* 2022; Riva *et al.* 2024) or assemblages of species (Belitz *et al.* 2025), rather than individual species. In terms of funding, positive discrimination mechanisms toward neglected taxa/habitats – or dedicated funding schemes – could be adopted (Mammola *et al.* 2020). Concerning publication practices, it is worth noting that scientific journals have already adopted statements on inclusivity – and one could make a case for the adoption of "taxonomic inclusivity statements" or similar measures. Such approach acknowledges also the influence of speciesism, where emotional resonance, symbolism, or perceived usefulness of species often dictate which organisms receive attention and which remain neglected (Jaquet 2025).

But probably the most critical dimension of addressing taxonomic bias and speciesism lies in improving how scientific results are communicated – especially when findings are incremental rather than sensational. Structural change will not occur if results documenting mild but persistent distortions remain confined to specialist audiences. However, effective communication is itself a specialized skill, and expecting researchers to shoulder the full burden of science communication is neither realistic nor equitable. Just as we rely on statisticians, data managers, and laboratory technicians, research teams should also include trained science communicators, designers, and outreach professionals – as exemplified by the comic illustration included in this paper (Figure 4). If biodiversity research is to influence societal and policy trajectories, institutions and funding bodies must recognize communication not as an optional add-on, but as an integral and professionally supported component of the research enterprise (Bickford *et al.* 2012; Meinard & Quétier 2014).

Without structural investment in communication expertise, even the most carefully documented biases risk remaining academically acknowledged but socially inert.

Data availability statement

Data and scripts supporting this study are permanently archived on Figshare (doi: [10.6084/m9.figshare.31095481](https://doi.org/10.6084/m9.figshare.31095481)) and are also available on GitHub to facilitate reproducibility of the analyses (link removed to ensure anonymity). (All links will be activated upon acceptance)

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TABLES AND FIGURE CAPTIONS

Table 1. Main independent variables used in this meta-analysis (Group), their granular subclassification (Subgroup), and examples of specific variables from the different studies

Group	Subgroup	Description	Example variables
Appearance	Attractiveness/ Charisma	Variables related to the overall attractiveness or charisma status of a species	Aesthetic, Salience score, Species appeal
	Size	Variables related to overall species body size	Body size, Body mass, Height, Flower diameter
	Unusual coloration	Proxy variables for unusual colorations (rare colors) or patterns (iridescence, multiple colors)	Blue and Red colorations, Overall colorfulness
	Common coloration	Proxy variables for more common colorations (relative to the taxon/geographic area)	Yellow, white, and light green colorations
	Unusual shape	Proxy variables for unusual morphological attributes or body shapes	Beak size, Tail length, crest length, Body asymmetry
	Cultural	Variables related to how culturally familiar or embedded a species is within human societies	Presence of a common name, Receiving a class on biodiversity, Number of search in Google,

Commonness			Year of description
	Ecological	Variables related to the ecological abundance or spatial and temporal frequency of species within their habitats	Flowering duration, Habitat breadth, Trophic level
	Geographical	Variables related to species range size, geographic distribution extent, and habitat breadth	Biogeographic realm, Latitudinal range, Range size
	Taxonomical	Variables related to taxonomic rarity or phylogenetic uniqueness of species	Evolutionary distinction, Number of congeneric species, Taxonomic uniqueness
Conservation	Priority	Variables indicating whether a species is designated as a conservation priority by organizations or policies	Presence/absence in conservation annexes, Species legal status
	Extinction risk	Variables related to the threat status or extinction probability of species	IUCN Red List category, Conservation status, Vulnerability score
General bias	Focal taxa versus control (6 macrotaxa considered)	Analyses comparing taxonomic groups (e.g., vertebrates vs. invertebrates, plants vs. animals) against control or reference groups	Taxonomic group (mammals, birds, reptiles, amphibians, fish, invertebrates), Kingdom-level comparisons
	Harmful	Variables related to species that	Whether the species

Utilitarianism		pose threats or dangers to humans	is a disease vector or a conflict species,
	Useful	Variables related to species that provide direct benefits or services to humans	Whether a species is edible, Ornamental species, Traded species
Other	Excluded from analysis	We grouped here any other difficult-to-classify predictor	Socio-economic status of the responder of a survey

FIGURE CAPTIONS

Figure 1. Overview of the 78 studies collected considering time, geography, taxonomy and assigned macro-variable categories. **A.** The temporal distribution of studies included in the meta-analysis; **B.** The worldwide distribution of studies, excluding those described as “global” ($n = 30$); **C.** Taxonomic distribution of 125 unique effects across the 78 studies; **D.** Distribution of 123 unique explanatory variables ([Table 1](#)) across the 78 studies.

Figure 2. Meta-regression of predicted bias of biodiversity knowledge and perception across the five main groups of independent variables (base models; A) and data subsets breaking down the effects by more fine scale independent variables (B). Effect sizes were measured as Fisher’s z and then back-transformed to Pearson’s r for visualisation; error bars represent the 95% confidence interval. Sample sizes for each moderator level are shown to the left of each estimate (number of studies | number of effect-size estimates). In A, individual background points are the single estimates for each study, with a density plot added to illustrate their general distribution.

Figure 3. Meta-regression of predicted bias of biodiversity knowledge and perception across the five main types of bias variables considered. Results are stratified by study methodology (**A**), dependent variable (**B**), geographic region (**C**), and taxonomic group (**D**). Effect sizes were measured as Fisher’s z and then backtransformed to Pearson’s r for visualisation; error bars represent the 95% confidence interval. Sample sizes for each moderator level are shown to the left of each estimate (number of studies | number of effect-size estimates). Vertical dashed lines denote no association ($r = 0$).

Figure 4. Comic illustrating the disparity in the taxonomic scope of biodiversity studies, reflecting speciesism, with vertebrates dominating the research landscape. This figure was

created by Jagoba Malumbres-Olarte, both a scientist and an artist with expertise in scientific communication and outreach, and originally appeared in a preprint (Mammola *et al.* 2022). Notably, the final published version of that paper (Mammola *et al.* 2023a) does not include it – a telling reminder that traditional peer review can still act as a filter against novel outreach formats. Original illustration by Jagoba Malumbres-Olarte.